

Supplementary Information: Recalibrating EuroLeague Performance Index Rating for outcome-concordant team ledgers

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ABSTRACT

Supplementary Information

This Supplementary Information contains extended methodological details, coefficient-uncertainty analyses, robustness and sensitivity analyses, and downstream team/player decompositions. It should be read together with the main manuscript.

Supplementary Methods

Coefficient uncertainty and sensitivity analyses

Supporting analyses quantify conditional coefficient uncertainty and test how the historical findings respond to alternative modelling and reporting choices. These analyses do not replace the season-locked primary construction.

Coefficient uncertainty was assessed with 500 whole-game bootstrap refits per target window. If $\mathcal{C}_Y^{*(b)}$ is a size- $|\mathcal{C}_Y|$ multiset drawn with replacement from $\mathcal{C}_Y$, let $q_{g,Y}^{*(b)} = \mathbf{x}_g^T \mathbf{E} \hat{\mathbf{a}}_Y^{*(b)}$ for each member of that multiset. Repeated games retain their sampling multiplicity. Replicate b was

$$\begin{aligned} \hat{\mathbf{a}}_Y^{*(b)} &= \hat{\mathbf{a}}_{\mathcal{C}_Y^{*(b)}}(\hat{\tau}_Y, \hat{\alpha}_Y), \\ \gamma_Y^{*(b)} &= \frac{\text{RMS}_{\mathcal{C}_Y^{*(b)}}(o)}{\text{RMS}_{\mathcal{C}_Y^{*(b)}}(q_Y^{*(b)})}, \\ \mathbf{c}_Y^{*(b)} &= \gamma_Y^{*(b)} \text{diag}(\mathbf{s}) \mathbf{E} \hat{\mathbf{a}}_Y^{*(b)}. \end{aligned} \tag{1}$$

The 2.5th and 97.5th empirical percentiles of these replicate quantities formed the conditional intervals shown in Figure 1. Hyperparameters remained fixed, so the intervals condition on the selection in Equation (main manuscript). The component residual analysis compared each official team field with the sum of corresponding player rows for all 4,036 team-games. Equations (main manuscript)–(main manuscript) were evaluated using unrounded coefficients.

Within the 2022–23 through 2024–25 calibration window used for the 2025–26 target, collinearity was described for both the ten-field signed matrix with rows $\mathbf{x}_g^T$ and the fitted eight-group matrix with rows $\mathbf{x}_g^T \mathbf{E}$. For each matrix, columns were centred and scaled to unit population standard deviation, and the maximum condition index was calculated as the ratio of the largest to the smallest singular value. Comparing the two indices distinguishes redundancy among the original ledger fields from conditioning of the grouped design actually used for estimation.

Retrospective sensitivity analyses altered the raw bounds to $[0.25, 1.75]$ and $[0.5, 2.0]$; used one- and two-season calibration windows; expanded both hyperparameter grids; replaced the single split by three forward-chaining cutoffs; excluded overtime; omitted each calibration season in turn; separated BLK from BLKA and FD from FC; excluded team-only residuals; removed the 68 completed 2021–22 games involving the three withdrawn Russian clubs; and restricted the 2025–26 evaluation to the regular season. The pace-normalised sensitivity used the conventional possession approximation²⁰

$$\hat{P}_{tg} = \text{FGA}_{tg} + 0.44 \text{FTA}_{tg} - \text{OREB}_{tg} + \text{TO}_{tg}, \quad \hat{P}_g = \frac{\hat{P}_{Wg,g} + \hat{P}_{Lg,g}}{2}, \tag{2}$$

where OREB denotes offensive rebounds. This was followed by the 70-possession transformation

$$\mathbf{x}_g^{(70)} = \frac{70}{\bar{P}_g} \mathbf{x}_g. \quad (3)$$

The value 0.44 is a conventional approximation rather than a EuroLeague-specific estimate; an empirical EuroLeague study estimated the corresponding coefficient as 0.454⁴⁷. Pace normalisation was therefore retained as a sensitivity analysis. The transformed design $\mathbf{x}_g^{(70)}$ replaced $\mathbf{x}_g$ only during inner selection, complete-window refitting and RMS anchoring for this sensitivity analysis. The resulting coefficient vector was then applied to the unnormalised target-season action ledger, as in the primary score.

The post-fit scale comparison considered six conventions. For a positive scale statistic T_A , the generic anchored factor was

$$\gamma_Y^{(A)} = \frac{T_A(\text{official calibration values})}{T_A(\text{fitted calibration values})}. \quad (4)$$

The ratio in Equation 4 was evaluated using winner–loser nonpoint contrasts for the RMS and population-SD conventions, and using official versus fitted nonpoint team totals for the team-total population SD, median absolute deviation about the median and interquartile-range conventions. No normalisation ($\gamma = 1$) was evaluated separately and is not an instance of that ratio. Both nonpoint and complete ordering, final coefficient envelopes, target-score dispersion and player-score/rank diagnostics were recomputed under every convention. Dependence on recurring clubs was examined by omitting each 2025–26 club in turn and by the club-membership and joint resampling procedures described above. Team rankings were compared between regular-season totals and all-stage per-game means.

Player scoring and secondary associations

Player scores use the locked team-calibrated row formula; no player-specific parameter is estimated. These analyses are therefore consequences of algebraic decomposability and secondary descriptive checks, not validation of individual player value.

Player RM-PIR applied Equation (main manuscript) with the 2025–26 locked coefficients to each official player row and then used Equation (main manuscript) to sum unrounded game scores by unique official player identifier. Players with no positive-minute appearance were excluded; players representing multiple clubs were aggregated once and all clubs were retained. Tied ranks followed the competition-ranking rule in Equation (main manuscript); names were ordered alphabetically for display only. The absence of player-, team- or coach-specific fitted parameters permits uniform calculation for entrants and transfers. It does not establish individual causal validity, independence from pace or role, or completeness of player evaluation. Calculations retained unrounded values throughout, with rounding applied only for presentation.

The secondary association analysis used all 8,741 positive-minute player-games and treated the game as the resampling cluster. Raw Spearman correlations related each score to official on-court plus-minus. Partial Spearman correlations first rank-transformed the score, plus-minus and one exposure covariate, residualised the first two rank vectors on the covariate rank and correlated the residuals; minutes and estimated possession exposure were evaluated separately, not simultaneously.

For this diagnostic only, let $\beta = 6532/15807 = 0.4132$, where 6,532 is the number of rule-classified possession-consuming free-throw trips and 15,807 is the number of recorded free-throw attempts in the 2025–26 play-by-play analysis. Dead-ball free-throw episodes were classified in the Game Center play-by-play records for all 402 games³⁷. Ordinary two- or three-attempt trips contributed one trip to the numerator; and-one attempts and technical, coach, bench, unsportsmanlike, disqualifying and other retained-possession awards were excluded. An ordinary trip whose final miss was recovered offensively still contributed one because the offensive-rebound term offsets the ensuing continuation in the possession expression. The classifier was deterministic and rule-based rather than independently hand-labelled. The play-by-play attempt totals reproduced the box-score FTA totals in all 804 team-games, establishing denominator completeness; this arithmetic agreement is not an independent validation of every episode classification. A 20,000-replicate whole-game bootstrap 95% percentile interval for β was 0.4097–0.4166; it conditions on the rule-based labels and does not include classification or feed uncertainty. Balanced game possessions and player exposure were

$$\begin{aligned} \tilde{P}_{tg} &= FGA_{tg} + \beta FTA_{tg} - OREB_{tg} + TO_{tg}, \\ \tilde{P}_g &= \frac{\tilde{P}_{Wg,g} + \tilde{P}_{Lg,g}}{2}, \\ \tilde{P}_{tg} &= \tilde{P}_g \frac{\text{MIN}_{ig}}{T_g}, \end{aligned} \quad (5)$$

where T_g is the elapsed regulation-plus-overtime duration of game g for one team. Aggregate team player-minutes equal $5T_g$ and are not the denominator in Equation 5. The coefficient β and the resulting exposure were used only in the secondary 2025–26 player diagnostics; neither entered RM-PIR estimation, RMS matching or scoring. Once estimated, β was applied to box scores from all 402 games. This event-derived coefficient answers a different measurement question from both the conventional 0.44 approximation and regression-based possession estimators^{20,47}.

Team-margin-residualised on-court plus-minus was defined as

$$U_{ig} = PM_{ig} - \frac{MIN_{ig}}{T_g} (PTS_{tg} - PTS_{\bar{t}g}), \quad (6)$$

where PM_{ig} is the player's observed on-court plus-minus and $\bar{t}$ denotes the opponent. The second term removes only the player's minutes-proportional share of the final team margin. This quantity is not adjusted plus-minus: it contains observed plus-minus and the final margin and includes no teammate, opponent, lineup or stint adjustment. A rate sensitivity correlated score per 100 estimated player possessions with $100U_{ig}/\tilde{P}_{ig}$ among the 7,342 rows with at least 18 estimated possessions. The exposure threshold reduces instability in very short appearances but does not remove the shared-denominator dependence. Finally, the current score was associated with U in the player's next appearance and the mean U over exactly the next three appearances. These analyses assess limited convergent association and temporal transport only; none validates individual credit or estimates causal impact.

Supplementary Results

Coefficient behaviour after nonpoint calibration

Only after the outcome comparison is established do the fitted coefficients show how the nonpoint ledger was reallocated. The fitted coefficient magnitudes varied across target seasons (Figure 1). For direct visual comparison, penalty coefficients are shown by absolute magnitude in the figure but retain negative signs in the scoring equation. Turnover penalties were consistently largest in absolute magnitude. In the 2025–26 calibration refits, the turnover group reached its upper raw bound in 100% of bootstrap samples and the paired block and foul groups reached their lower raw bounds in 100%. In the corresponding calibration window, the standardised ten-field signed design had a maximum condition index of 100.79, whereas the fitted eight-group design had an index of 3.44. The contrast shows that the severe pre-grouping redundancy was substantially removed by the block and foul equality restrictions; it does not support describing the fitted design itself as ill-conditioned. Boundary activity nevertheless precludes interpreting a coefficient as an isolated marginal or causal event value. The vector is one constrained scoring direction, and its displayed percentile ranges condition on the selected hyperparameters.

RMS matching as a scale safeguard

For 2025–26, the inner selection procedure selected $\tau = 5$ and $\alpha = 0.01$ from the 2022–23 through 2024–25 calibration window. The complete-window refit gave $\gamma_T = 1.0205$. Although raw group magnitudes were constrained to $[0.5, 1.5]$, the final magnitudes ranged from 0.510 to 1.531 after scaling. The official and rescaled fitted nonpoint RMS values were both 19.652 by construction. This equality concerns calibration-sample winner–loser contrasts only; it does not imply equality of centred variance, means, target-season scale, player-score variance or complete-score distributions. Because $\gamma_T > 0$, it cannot change nonpoint signs. Across six scale choices, including no post-fit normalisation, the 2025–26 nonpoint count was therefore invariant at 357 winner-higher games; complete-score counts ranged from 380 to 382, compared with 381 under RMS matching. RMS matching functions here as an interpretive scale safeguard, not as the empirical source of the observed reclassifications.

Robustness, sensitivity and numerical stability

The principal findings were complemented by robustness, sensitivity and numerical-stability analyses.

The 2025–26 nonpoint change remained negative across 16 retrospective sensitivities covering exact alternative raw bounds $[0.25, 1.75]$ and $[0.5, 2.0]$, calibration windows, hyperparameter grids and cutoffs, overtime exclusion, pace normalisation, equality restrictions, scale anchors, calibration-season omissions, team-only residual exclusion and regular-season-only evaluation. Complete-score changes were smaller and more specification-sensitive. Leave-one-club-out changes ranged from -6.70 to -4.67 nonpoint percentage points and from -2.20 to -1.10 complete-score points. Nine, 6, 3, 16 and 3 hyperparameter-grid candidates tied for maximum inner concordance across the 2021–22 through 2025–26 targets. Refitting every tied candidate preserved the direction of both effects; in 2025–26 all three tied candidates reproduced exactly 357 nonpoint and 381 complete winner-higher games. The deterministic tie-breaking rule favours stronger shrinkage, then smaller τ , then lower soft loss among count-equivalent candidates.

Excluding the 68 completed 2021–22 games involving CSKA Moscow, UNICS Kazan or Zenit St Petersburg and repeating selection, refitting and RMS matching preserved the direction of the RM-PIR-minus-PIR change in every affected target;

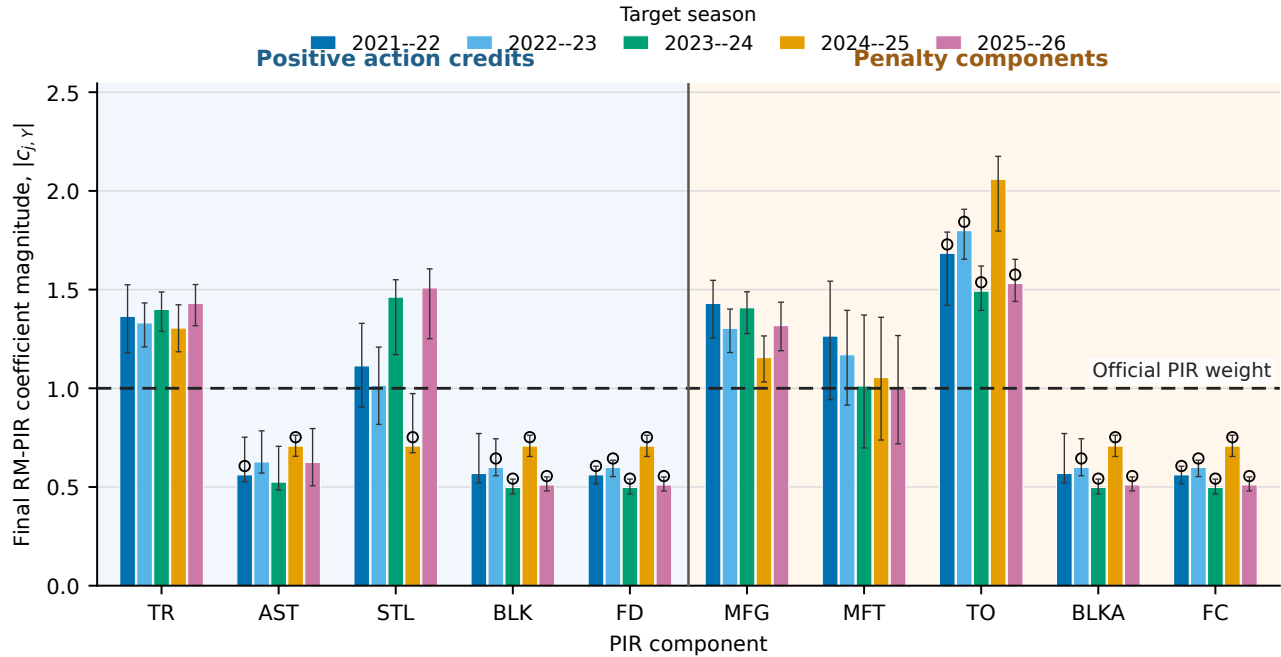


Figure 1. Season-locked RM-PIR coefficient magnitudes by target season. Bars to the left of the vertical separator represent positive action credits; bars to the right represent penalty magnitudes, although these components enter the RM-PIR score with negative signs. Thin error bars show conditional 95% percentile intervals from whole-game bootstrap refits. The dashed horizontal line at one denotes the absolute unit coefficient used by official PIR (+1 for credited actions and −1 for penalised actions). PTS is not displayed because its coefficient remains fixed at +1 in both indices. Open circles identify components whose corresponding raw action-group magnitude reached an imposed bound. These bounds apply before RMS scaling and therefore do not constrain the displayed final magnitudes to $[0.5, 1.5]$. BLK and BLKA, and FD and FC, have equal magnitudes by construction.

the reduced 2021–22 target contained 231 games. For 2021–22 through 2024–25, complete-score changes were −2.60, −3.35, −1.81 and −4.85 percentage points, respectively; nonpoint changes were −5.63, −3.96, −5.14 and −6.06 points. The 2025–26 result was unchanged because its calibration window begins in 2022–23. Thus the reported direction does not depend on treating the withdrawn-club games as part of the completed-game population.

The vector fitted only to 2023–24 and transported retrospectively to 2024–25 and 2025–26 yielded complete-score counts of 315/330 and 383/402, respectively, compared with official counts of 298 and 375. Those gains were 17 and 8 games, with no newly introduced discordance. This fixed-origin experiment is a sensitivity analysis, not the operational annual rule: 2024–25 was available when the 2025–26 vector was locked, so the rolling calibration using 2022–23 through 2024–25 yielded 381/402. The 2025–26 season is discussed because it is the latest complete season, not because it had the largest transported gain.

All inner-grid and conditional coefficient-bootstrap fits converged. Coefficients from three dispersed starting values differed by at most 5.06×10^{-7} ; at the unit starting vector in the first calibration window, the Euclidean discrepancy between the analytic and finite-difference gradients was 8.91×10^{-9} ; and the maximum RMS identity error was 3.55×10^{-15} . These checks establish numerical reproducibility, not substantive coefficient stability or causal interpretation. Across the six 2025–26 scale conventions, player-game negative-score rates ranged from 14.91% to 16.38%, season-rank correlations with the RMS version were at least 0.9992 and 19–20 of its top 20 players were retained. Scale choice therefore changed score dispersion and some ranks even when nonpoint team ordering was invariant.

Team and player consequences of ledger reallocation

The remaining analyses describe downstream consequences of applying the same locked coefficients to the recorded ledger. They are not individual-player validation and do not convert algebraic decomposition into causal contribution.

For teams, a common-stage comparison was used to avoid unequal postseason exposure. Table 1 contains all 20 clubs over the 38-game 2025–26 regular season. Because every club played the same number of games, total and per-game rankings coincide. Dubai recorded the largest upward movement, whereas Maccabi recorded the largest downward movement.

Table 1. Supplementary Table S1. 2025–26 regular-season team rankings. GP denotes games played. Every club played 38 games, so ranking by total or per-game mean is identical. Rank change is official PIR rank minus RM-PIR rank; positive values indicate upward movement. RM-PIR used coefficients selected and fitted exclusively from 2022–23 through 2024–25.

Team	Code	GP	PIR total	RM total	PIR/game	RM/game	PIR rank	RM rank	Δ rank
Olympiacos	OLY	38	4164	3859.92	109.58	101.58	1	1	0
Monaco	MCO	38	4036	3773.86	106.21	99.31	2	2	0
Real	MAD	38	3957	3724.25	104.13	98.01	3	3	0
Valencia	PAM	38	3921	3675.93	103.18	96.74	4	4	0
Zalgiris	ZAL	38	3771	3547.25	99.24	93.35	6	5	+1
Dubai	DUB	38	3737	3545.84	98.34	93.31	9	6	+3
Maccabi	TEL	38	3803	3539.55	100.08	93.15	5	7	-2
Panathinaikos	PAN	38	3769	3535.72	99.18	93.05	7	8	-1
Hapoel Tel Aviv	HTA	38	3753	3528.64	98.76	92.86	8	9	-1
Crvena Zvezda	RED	38	3714	3480.16	97.74	91.58	10	10	0
Milan	MIL	38	3671	3409.86	96.61	89.73	11	11	0
Baskonia	BAS	38	3626	3362.99	95.42	88.50	12	12	0
Paris	PRS	38	3531	3344.34	92.92	88.01	14	13	+1
Barcelona	BAR	38	3558	3342.08	93.63	87.95	13	14	-1
Virtus	VIR	38	3376	3136.22	88.84	82.53	15	15	0
Fenerbahce	ULK	38	3283	3105.13	86.39	81.71	17	16	+1
Partizan	PAR	38	3287	3094.08	86.50	81.42	16	17	-1
Bayern	MUN	38	3260	3053.61	85.79	80.36	18	18	0
Anadolu Efes	IST	38	3239	2992.09	85.24	78.74	19	19	0
ASVEL	ASV	38	3052	2825.16	80.32	74.35	20	20	0

Supplementary Figure 2 displays the complete rank transition.

Table 2 extends the same scoring comparison to every completed stage in 2025–26. Its cumulative ranks are based on season totals, whereas the adjacent per-game columns show scoring rate. Postseason qualification created unequal exposure, so these all-stage cumulative ranks describe both performance and opportunity and are not equal-opportunity alternatives to the primary regular-season ranks.

Official team totals need not equal the sum of player rows because official team rows may contain unallocated events. The points residual was zero in every team-game. After explicitly weighting each component residual, the identity in Equation (main manuscript) held to a maximum absolute error of 4.22×10^{-14} . Replacing 2025–26 official team rows with player sums changed the complete-score classification in 7 games; the analogous RM-PIR substitution changed 13. Decomposability is exact only when the explicit team-only residual is retained, and it does not justify assigning that residual to an individual.

The 2025–26 coefficients were applied without player- or team-specific parameters. Table 3 reports all 335 players with a positive-minute appearance, combining observations for transferred players under their unique official player identifiers and listing all represented clubs. The rankings use all-stage season totals, so games played and minutes are reported to make exposure explicit. Rank changes describe the redistribution of recorded action credit; they do not establish a change in causal player impact.

Table 3. Supplementary Table S3. All 335 players with a positive-minute appearance in 2025–26. Official PIR and RM-PIR are all-stage season totals; the RM-PIR coefficients were estimated from 2022–23 through 2024–25 and locked before the target season. GP denotes games played and MIN total minutes. Ranks use competition ranking on season totals. Rank change is official rank minus RM-PIR rank, so positive values indicate upward movement. Team(s) are official club codes and list every club represented by a transferred player during 2025–26.

Player	Team(s)	GP	MIN	PIR	RM-PIR	PIR rank	RM rank	Δ rank
Vezenkov, Sasha	OLY	39	1086	863	829.46	1	1	0
Milutinov, Nikola	OLY	40	861	778	788.45	3	2	+1
Oturu, Dan	HTA	42	1004	704	721.64	7	3	+4
Wright, Moses	ZAL	42	993	720	718.46	6	4	+2

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Player ranking table continued

Player	Team(s)	GP	MIN	PIR	RM-PIR	PIR rank	RM rank	Δ rank
Bryant, Elijah	HTA	38	1120	744	643.79	4	5	-1
Moneke, Chima	RED	35	942	648	635.59	11	6	+5
Kabengele, Mfiondu	DUB	36	801	625	630.20	14	7	+7
Tavares, Walter	MAD	39	888	603	619.25	15	8	+7
Hernangomez, Juancho	PAN	44	1076	542	607.55	27	9	+18
Hoard, Jaylen	TEL	35	1008	565	596.92	24	10	+14
Lyles, Trey	MAD	42	890	634	588.86	13	11	+2
Francisco, Sylvain	ZAL	42	1167	806	575.64	2	12	-10
Petrusev, Filip	DUB	38	893	584	556.98	21	13	+8
Diallo, Alpha	MCO	42	1172	572	556.95	23	14	+9
James, Mike	MCO	39	1168	743	545.41	5	15	-10
Tubelis, Azuolas	ZAL	42	953	535	541.79	31	16	+15
Bonga, Isaac	PAR	38	1056	544	534.58	25	17	+8
Montero, Jean	PAM	38	874	679	523.92	9	18	-9
Taylor, Kamaron	PAM	40	962	542	512.48	27	19	+8
Luwawu-Cabarrot, Timothe	BAS	36	915	590	509.10	18	20	-2
Baldwin IV, Wade	ULK	42	1086	689	497.06	8	21	-13
Sorkin, Roman	TEL	38	870	536	496.26	30	22	+8
Theis, Daniel	MCO	38	780	462	488.57	42	23	+19
Reuvers, Nathan	PAM	44	838	482	488.32	39	24	+15
Horton Tucker, Talen	ULK	41	977	596	483.00	17	25	-8
Wright IV, McKinley	DUB	38	1123	597	481.17	16	26	-10
Hall, Donta	OLY	38	580	436	475.89	47	27	+20
Shengelia, Tornike	BAR	34	848	522	474.50	32	28	+4
Osman, Cedi	PAN	39	1065	496	467.83	35	29	+6
Campazzo, Facundo	MAD	44	1073	635	467.81	12	30	-18
Nwora, Jordan	RED	34	922	541	466.94	29	31	-2
Blossomgame, Jaron	MCO	43	1020	453	464.03	43	32	+11
Hezonja, Mario	MAD	44	995	544	457.15	25	33	-8
Jones, Tyrique	OLY; PAR	33	641	434	452.40	49	34	+15
Hifi, Nadir	PRS	37	824	650	450.35	10	35	-25
Nunn, Kendrick	PAN	37	1100	579	450.28	22	36	-14
Niang, Saliou	VIR	38	823	415	445.68	54	37	+17
Dorsey, Tyler	OLY	43	1082	588	444.12	19	38	-19
Miller-McIntyre, Codi	RED	39	1206	585	443.30	20	39	-19
Leday, Zach	MIL	30	739	452	418.67	44	40	+4
Clyburn, Will	BAR	33	844	485	417.59	38	41	-3
Pradilla, Jaime	PAM	44	858	409	410.09	55	42	+13
Brissett, O'Shae J	TEL	36	885	376	403.87	69	43	+26
Punter, Kevin	BAR	39	1060	517	399.04	33	44	-11
Mirotic, Nikola	MCO	31	591	393	390.39	65	45	+20
Jones, Chris	HTA	42	859	496	388.03	35	46	-11
Brooks, Armoni	MIL	38	857	400	386.39	62	47	+15
Izundu, Ebuka	RED	33	475	341	384.57	89	48	+41
Osmani, Ercan	IST	36	880	409	383.53	55	49	+6
Melli, Nicolo	ULK	35	831	354	378.23	78	50	+28
Diakite, Mamadi	BAS	37	713	375	377.62	71	51	+20
Okeke, Chuma	MAD	43	770	342	376.81	88	52	+36
Shields, Shavon	MIL	35	941	448	376.28	46	53	-7
Okobo, Elie	MCO	42	1074	493	372.52	37	54	-17
Deck, Gabriel	MAD	41	837	358	371.68	75	55	+20
Fernando, Bruno	MAD; PAR	37	641	361	370.72	74	56	+18
Thompson, Darius	PAM	44	829	435	370.15	48	57	-9
Vesely, Jan	BAR	38	804	358	368.70	75	58	+17
Peters, Alec	OLY	43	653	365	367.90	73	59	+14
Nebo, Josh	MIL	26	560	357	367.68	77	60	+17
Blakeney, Antonio	HTA	41	929	429	367.42	50	61	-11
Malcolm, Collin	HTA	40	996	340	366.88	92	62	+30
Sloukas, Kostas	PAN	38	810	477	364.69	40	63	-23
Satoransky, Tomas	BAR	37	922	405	359.63	60	64	-4
Strazel, Matthew	MCO	43	1046	475	353.20	41	65	-24
Dokossi, Allan	PRS	34	550	293	352.80	107	66	+41
Robinson, Justin	PRS	36	683	517	351.99	33	67	-34
Weiler-Babb, Nick	IST	37	1066	406	351.75	58	68	-10
Colson, Bonzie	ULK	40	675	347	348.91	84	69	+15
Maledon, Theo	MAD	38	647	426	341.18	51	70	-19
Mike, Isiaha	MUN	37	828	339	340.18	94	71	+23
Diouf, Mouhamet	VIR	33	612	345	339.11	86	72	+14
Brown, Sterling	PAR	37	1041	408	337.08	57	73	-16
Lucic, Vladimir	MUN	33	765	352	335.25	81	74	+7
Ellis, Quinn	MIL	36	912	406	334.97	58	75	-17
Faye, Mouhamed	PRS	38	653	268	334.73	119	76	+43
Hayes, Kevarius	MCO	40	697	290	333.32	109	77	+32
Ndiaye, Mbaye	ASV	32	757	328	332.96	98	78	+20
Morgan, Matthew	VIR	34	678	416	325.54	53	79	-26
Biberovic, Tarik	ULK	41	998	341	323.14	89	80	+9
Micic, Vasilije	HTA	40	964	450	322.66	45	81	-36
Booker, Devin	MIL	36	718	345	322.44	86	82	+4
Costello, Matthew	PAM	40	662	295	318.31	105	83	+22

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Player ranking table continued

Player	Team(s)	GP	MIN	PIR	RM-PIR	PIR rank	RM rank	Δ rank
Ojeleye, Semi	RED	39	894	306	318.27	103	84	+19
Leaf, TJ	TEL	24	437	301	317.84	104	85	+19
Bacon, Dwayne	DUB	29	852	404	313.83	61	86	-25
Jones, Kai	IST	36	557	266	312.39	121	87	+34
Forrest, Trent	BAS	27	645	392	312.36	66	88	-22
Butkevicius, Arnas	ZAL	38	685	289	311.42	110	89	+21
De Colo, Nando	ASV; ULK	30	564	400	310.50	62	90	-28
Watson, Glynn	ASV	30	767	376	310.19	69	91	-22
Guduric, Marko	MIL	35	771	390	309.80	67	92	-25
Obst, Andreas	MUN	35	931	421	305.66	52	93	-41
Badio, Brancou	PAM	40	820	383	305.21	68	94	-26
Da Silva, Oscar	MUN	35	557	271	304.16	118	95	+23
Parra, Joel	BAR	40	746	274	301.53	115	96	+19
Alston Jr., Derrick	VIR	37	830	322	294.92	99	97	+2
Grant, Jerian	PAN	42	993	349	293.96	83	98	-15
Loyd, Jordan	IST	34	798	366	291.55	72	99	-27
Jessup, Justinian	MUN	37	715	291	291.19	108	100	+8
Garuba, Usman	MAD	40	500	256	289.52	127	101	+26
Sleva, Dustin	ZAL	42	769	253	289.43	131	102	+29
Faried, Kenneth	PAN	35	460	264	288.59	122	103	+19
Sako, Neal	PAM	42	562	259	288.03	125	104	+21
Walkup, Thomas	OLY	37	889	353	283.64	80	105	-25
Hernangomez, Willy	BAR	39	528	273	282.16	116	106	+10
Osetkowski, Dylan	PAR	35	703	255	279.43	128	107	+21
Birch, Khem	ULK	43	711	210	278.81	149	108	+41
Williams-Goss, Nigel	ZAL	31	746	340	274.91	92	109	-17
Bolmaro, Leandro	MIL	29	655	309	271.98	101	110	-9
Motley, Johnathan	HTA	35	512	283	270.33	112	111	+1
Ulanovas, Edgaras	ZAL	42	779	279	265.76	114	112	+2
Fournier, Evan	OLY	39	855	351	263.25	82	113	-31
Walker IV, Lonnie	TEL	28	637	354	260.79	78	114	-36
Vildoza, Luca	VIR	31	660	307	260.76	102	115	-13
Simmons, Kobi	BAS	31	688	336	260.41	95	116	-21
Moore, Omari	PAM	43	742	334	257.11	97	117	-20
Butler, Jared	RED	32	640	346	256.96	85	118	-33
Feliz, Andres	MAD	39	649	262	256.58	123	119	+4
Traore, Armel	ASV	36	572	223	254.39	137	120	+17
Shorts, TJ	PAN	43	693	341	252.80	89	121	-32
Ward, Tyson	OLY	35	668	272	244.30	117	122	-5
Kalinic, Nikola	RED	39	903	243	243.21	133	123	+10
Key, Braxton	PAM	34	579	225	239.82	136	124	+12
Edwards, Carsen	VIR	36	915	395	238.32	64	125	-61
Musa, Dzanan	DUB	18	480	281	237.10	113	126	-13
Willis, Derek	PRS	28	454	223	236.69	137	127	+10
Birutis, Laurynas	ZAL	41	410	215	235.37	144	128	+16
Lo, Maodo	ZAL	41	739	288	232.15	111	129	-18
Clark III, Jimmy	TEL	37	730	318	230.43	100	130	-30
Harrison, Shaquille	ASV	29	549	258	230.11	126	131	-5
Smits, Rolands	IST	35	664	211	229.74	147	132	+15
Anderson, Justin	DUB	33	775	209	228.41	151	133	+18
Hall, Devon	ULK	35	884	295	227.70	105	134	-29
Cordinier, Isaia	IST	30	726	260	225.31	124	135	-11
McCormack, David	MUN	28	433	214	223.49	146	136	+10
Gabriel, Wenyen	MUN	36	610	204	221.97	154	137	+17
Ricci, Giampaolo	MIL	38	624	168	219.96	176	138	+38
Wainright, Ish	HTA	42	968	176	216.24	170	139	+31
Diop, Khalifa	BAS	28	469	172	216.03	172	140	+32
Seljaas, Zachary	ASV	36	728	211	215.69	147	141	+6
Smailagic, Alen	VIR	31	511	221	214.97	141	142	-1
Frisch, Clement	BAS	37	449	179	214.10	168	143	+25
Puerto, Josep	PAM	38	598	199	212.10	156	144	+12
Omoruyi, Eugene	BAS	23	465	255	211.76	128	145	-17
Angola, Braian	ASV	15	366	268	211.03	119	146	-27
Vautier, Bastien	ASV	38	482	194	209.63	159	147	+12
Santos, Marcio	TEL	37	520	221	206.39	141	148	-7
Jantunen, Mikael	ULK	36	669	178	204.66	169	149	+20
Norris, Miles	BAR	36	446	169	203.12	175	150	+25
Cavaliere, Leopold	PRS	36	504	164	201.69	180	151	+29
Diarra, Aliou	VIR	28	298	180	200.06	166	152	+14
Blatt, Tamir	TEL	38	738	336	199.29	95	153	-58
Jallow, Karim	VIR	37	777	202	198.33	155	154	+1
Jones, Carlik	PAR	15	421	251	197.15	132	155	-23
Spagnolo, Matteo	BAS	37	572	232	197.09	134	156	-22
Morgan, Jeremy	PRS	32	567	185	193.40	160	157	+3
Holmes, Richaun	PAN	19	356	172	192.64	172	158	+14
Kurucs, Rodions	BAS	26	550	184	190.09	162	159	+3
Hayes-Davis, Nigel	PAN	16	457	206	185.03	153	160	-7
Rhoden, Jared	PRS	26	555	208	184.78	152	161	-9
Kamenjas, Kenan	DUB	28	304	162	183.13	181	162	+19

Continued on next page

Player ranking table continued

Player	Team(s)	GP	MIN	PIR	RM-PIR	PIR rank	RM rank	Δ rank
Brizuela, Dario	BAR	38	591	223	182.96	137	163	-26
Washington, Duane	PAR	23	504	254	181.83	130	164	-34
Massa, Bodian	ASV	34	539	151	181.59	191	165	+26
Avramovic, Aleksa	DUB	22	374	218	177.36	143	166	-23
Stevens, Lamar	PRS	23	459	183	175.07	164	167	-3
M'Baye, Amath	PRS	34	588	210	174.83	149	168	-19
Mitoglou, Konstantinos	PAN	31	401	166	173.59	177	169	+8
Lessort, Mathias	PAN	14	317	170	172.72	174	170	+4
Motiejunas, Donatas	RED	27	339	198	172.35	157	171	-14
Yurtseven, Omer	PAN	19	240	159	170.50	184	172	+12
Herrera, Sebastian	PRS	36	591	180	168.91	166	173	-7
Rogkavopoulos, Nikolaos	PAN	34	419	158	168.23	186	174	+12
Abalde, Alberto	MAD	42	654	158	164.19	186	175	+11
Dimitrijevic, Neno	MUN	16	334	223	163.11	137	176	-39
Jekiri, Tonye	PAR	15	289	148	162.79	192	177	+15
Nedovic, Nemanja	MCO	35	470	174	162.65	171	178	-7
Lee, Saben	IST; OLY	27	453	215	157.75	144	179	-35
Poirier, Vincent	IST	14	259	154	155.79	188	180	+8
Hackett, Daniel	VIR	34	499	153	151.05	190	181	+9
Pajola, Alessandro	VIR	24	458	165	149.53	178	182	-4
Calathes, Nick	PAR	31	596	184	148.56	162	183	-21
Eboua, Paul	ASV	18	338	135	147.47	199	184	+15
Tarpey, Terry	MCO	36	277	118	145.41	209	185	+24
Voigtman, Johannes	MUN	23	299	145	145.35	193	186	+7
Bertans, Davis	DUB	29	541	160	144.97	183	187	-4
Lundberg, Iffe	TEL	16	390	198	141.62	157	188	-31
Radzevicius, Gytis	BAS	25	443	126	140.54	205	189	+16
Diallo, Hamidou	BAS	13	288	161	138.97	182	190	-8
DiBartolomeo, John	TEL	27	387	121	136.94	207	191	+16
Dozier, PJ	IST	26	571	185	135.74	160	192	-32
Rayman, William	TEL	38	455	105	135.71	220	193	+27
Sedekerskis, Tadas	BAS	13	310	130	132.22	203	194	+9
Larkin, Shane	IST	12	364	181	129.18	165	195	-30
Odiase, Tai	HTA	29	346	114	128.81	212	196	+16
Bolombo, Joel	RED	15	217	114	128.69	212	197	+15
Cale, Myles	BAR	34	516	110	128.03	215	198	+17
Lighty, David	ASV	33	613	130	126.92	203	199	+4
Ouattara, Yakuba	PRS	34	573	144	126.62	196	200	-4
Heurtel, Thomas	ASV	25	487	226	126.53	135	201	-66
Hollatz, Justus	MUN	26	433	122	126.28	206	202	+4
Swider, Cole	IST	30	270	117	125.40	211	203	+8
Nowell, Markquis	BAS	28	403	165	123.68	178	204	-26
McKissic, Shaquille	OLY	27	312	135	122.63	199	205	-6
Giffey, Niels	MUN	32	423	118	121.63	209	206	+3
Dobric, Ognjen	RED	25	313	112	121.38	214	207	+7
Jovic, Stefan	MUN	19	351	145	118.08	193	208	-15
Howard, Markus	BAS	23	475	159	116.69	184	209	-25
Boston Jr., Brandon	ULK	28	331	107	114.36	219	210	+9
Akele, Nicola	VIR	27	267	86	110.39	228	211	+17
Brazdeikis, Ignas	ZAL	38	562	119	108.81	208	212	-4
De Larrea, Sergio	PAM	31	346	144	108.05	196	213	-17
Dessert, Brice	IST	17	220	93	104.73	224	214	+10
Milton, Shake	PAR	23	427	136	98.70	198	215	-17
Hombres, Daulton	PRS	21	285	90	97.67	225	216	+9
Begarín, Juhann	MCO	23	310	86	96.46	228	217	+11
Fall, Youssoupha	BAR	18	135	86	95.05	228	218	+10
Downton Jr, Jeffrey	TEL	19	326	132	94.97	201	219	-18
Rathan-Mayes, Xavier	MUN	32	487	154	94.74	188	220	-32
Pokusevski, Aleksej	PAR	27	326	78	94.52	236	221	+15
Laprovittola, Nicolas	BAR	20	337	145	94.41	193	222	-29
Davidovac, Dejan	RED	21	299	73	94.40	240	223	+17
Ajinca, Melvin	ASV	31	549	105	93.79	220	224	-4
Beaubois, Rodrigue	IST	21	261	98	91.67	222	225	-3
Caboclo, Bruno	DUB	11	140	77	90.63	237	226	+11
Abass, Awudu	DUB	31	508	73	89.95	240	227	+13
Dunston, Bryant	MIL	17	186	80	89.69	234	228	+6
Hazer, Sehmus	IST	33	435	131	89.19	202	229	-27
Len, Alex	MAD	23	211	85	87.84	231	230	+1
Joseph, Cory	OLY	16	249	108	87.02	216	231	-15
Lacic, Arijan	PAR	25	328	68	85.73	245	232	+13
Edwards, Kessler	HTA	13	171	73	85.23	240	233	+7
Bitim, Onuralp	ULK	29	308	70	83.66	243	234	+9
Birsén, Metecan	ULK	24	270	64	82.85	247	235	+12
Payne, Cameron	PAR	10	233	108	82.21	216	236	-20
Sirvydis, Deividas	ZAL	27	331	95	80.42	223	237	-14
Prepelic, Klemen	DUB	27	363	90	80.19	225	238	-13
Papanikolaou, Kostas	OLY	36	515	82	78.11	232	239	-7
Marinkovic, Vanja	PAR	19	389	79	71.85	235	240	-5
Silva, Chris	ULK	16	139	68	69.50	245	241	+4

Continued on next page

Player ranking table continued

Player	Team(s)	GP	MIN	PIR	RM-PIR	PIR rank	RM rank	Δ rank
Parker, Jabari	PAR	15	289	76	67.86	238	242	-4
Dinwiddie, Spencer	MUN	11	254	108	65.40	216	243	-27
Kondic, Kosta	DUB	16	188	60	63.76	249	244	+5
Brown, Lorenzo	MIL	12	174	82	63.76	232	245	-13
Lavi, Gur	TEL	15	169	50	58.57	251	246	+5
Hankins, Zachary	TEL	10	103	46	56.22	253	247	+6
Jackson, Edwin	ASV	21	330	90	55.40	225	248	-23
Yilmaz, Erkan	IST	23	204	38	53.31	259	249	+10
Ginat, Tomer	HTA	14	137	43	47.64	254	250	+4
Bacot Jr., Armando	ULK	26	263	35	46.68	265	251	+14
Ferrari, Francesco	VIR	12	115	41	44.23	257	252	+5
Ntilikina, Frank	OLY	24	394	62	43.86	248	253	-5
Ayayi, Joel	PRS	13	137	35	43.04	265	254	+11
Rivero, Jasiel	RED	9	88	39	41.99	258	255	+3
Baldwin, Kamar	MUN	9	147	50	41.28	251	256	-5
Giedraitis, Dovydas	ZAL	20	250	36	40.73	262	257	+5
Ellis, Boogie	DUB	5	89	43	39.95	254	258	-4
Madar, Yam	HTA	16	155	51	39.39	250	259	-9
Makoundou, Yoan	MCO	5	37	38	39.09	259	260	-1
Marcos, Juani	BAR	31	266	43	36.38	254	261	-7
Llull, Sergio	MAD	40	426	76	35.40	238	262	-24
Villar, Rafa	BAS	27	168	25	35.16	273	263	+10
Shahrvin, Enzo	PRS	11	95	24	34.86	274	264	+10
Samodurov, Alexandros	PAN	13	80	36	33.94	262	265	-3
Kalaitzakis, Panagiotis	PAN	26	133	26	33.29	271	266	+5
Samanic, Luka	BAS	5	69	36	31.63	262	267	-5
Mannion, Niccolo	MIL	22	254	70	29.63	243	268	-25
Ennis, Tyler	HTA	10	84	32	28.10	269	269	0
Wilbekin, Scottie	ULK	6	89	35	28.02	265	270	-5
Bako, Ismael	PRS	9	83	23	27.22	275	271	+4
Papagiannis, Georgios	IST	5	74	26	26.75	271	272	-1
Miljenovic, Stefan	RED	13	90	28	25.68	270	273	-3
Toliopoulos, Vassilis	PAN	10	96	34	24.52	268	274	-6
Sestina, Nathan	MIL; PAM	13	110	20	24.20	279	275	+4
Timor, Bar	HTA	10	58	18	21.56	280	276	+4
Dangubic, Nemanja	DUB	14	107	8	20.26	293	277	+16
Tonut, Stefano	MIL	10	90	14	19.85	283	278	+5
Keita, Sayon	BAR	6	52	15	17.79	281	279	+2
Rubstavicius, Mantas	ZAL	17	73	15	16.38	281	280	+1
Kramer, David	MAD	14	159	12	16.32	284	281	+3
Sanli, Sertac	DUB	7	55	10	14.39	288	282	+6
Randolph, Levi	HTA	12	123	8	12.16	293	283	+10
Zagars, Arturs	ULK	5	56	21	12.14	277	284	-7
Carter, Tyson	RED	11	162	22	11.74	276	285	-9
Taylor, Brandon	VIR	11	83	21	11.66	277	286	-9
Grigonis, Marius	PAN	19	111	9	11.14	290	287	+3
Tote, Leonardo	MIL	1	17	11	11.01	285	288	-3
Nogues, Isaac	PAM	13	42	4	10.73	301	289	+12
Antetokounmpo, Kostas	OLY	4	23	9	10.71	290	290	0
Mutaf, David	IST	8	50	11	9.02	285	291	-6
Fall, Moustapha	OLY	2	13	11	8.94	285	292	-7
Procida, Gabriele	MAD	8	40	7	7.87	296	293	+3
Sima, Yankuba	PAM	6	49	3	7.75	304	294	+10
Plavsic, Uros	RED	7	36	4	7.36	301	295	+6
Mikalauskas, Kajus	ZAL	2	18	9	7.09	290	296	-6
Accorsi, Matteo	VIR	6	25	3	5.30	304	297	+7
Diop, Ousmane	MIL	2	13	6	5.19	297	298	-1
Morris, Monte	OLY	8	92	10	5.11	288	299	-11
Radanov, Aleksa	PAR	10	68	2	5.07	310	300	+10
Kratzer, Leon	MUN	9	58	-4	4.67	329	301	+28
Omoruyi, Clifford	TEL	3	16	3	4.48	304	302	+2
Flaccadori, Diego	MIL	7	41	6	4.12	297	303	-6
Palatin, Guy	HTA	10	22	1	3.85	312	304	+8
Nakic, Mario	PAR	1	6	3	3.43	304	305	-1
Ryan, Matt	DUB	2	13	4	2.61	301	306	-5
Iroegbu, Ike	PAM	2	25	5	2.08	300	307	-7
Bol, Juom Maker	BAS	1	3	2	1.80	310	308	+2
Lopez-Arostegui, Xabi	PAM	10	85	3	1.41	304	309	-5
Hrabar, Vit	BAS	4	10	0	1.21	315	310	+5
Bosnjakovic, Mitar	PAR	3	9	0	0.92	315	311	+4
Kouzeloglou, Ioannis	PAN	3	17	0	0.81	315	312	+3
Trifunovic, Uros	TEL	1	3	1	0.68	312	313	-1
Dos Santos, Yago	RED	25	269	38	0.36	259	314	-55
Evans, Keenan	OLY	1	2	0	0.00	315	315	0
Grinvalds, Gunars	MAD	1	1	0	0.00	315	315	0
Radosic, Ognjen	RED	1	1	0	0.00	315	315	0
Andjusic, Danilo	DUB	1	4	0	-0.64	315	318	-3
Armstrong, Taran	DUB	7	40	-4	-0.76	329	319	+10
Muurinen, Miikka	PAR	6	11	0	-0.80	315	320	-5

Continued on next page

Player ranking table continued

Player	Team(s)	GP	MIN	PIR	RM-PIR	PIR rank	RM rank	Δ rank
Mahmutoglu, Melih	ULK	7	42	1	-1.00	312	321	-9
Almansa, Izan	MAD	3	10	-2	-1.06	325	322	+3
Kusturica, Nikola	BAR	1	1	-2	-1.83	325	323	+2
Segev, Itay	HTA	1	3	-2	-1.83	325	323	+2
Graham, Devonte	RED	7	80	6	-2.10	297	325	-28
Ngouama, Mehdy	ASV	12	129	3	-2.28	304	326	-22
Netzipoglou, Omiros	OLY	4	15	-4	-2.45	329	327	+2
Michineau, David	MCO	4	9	-1	-2.77	324	328	-4
Larentzakis, Giannoulis	OLY	7	21	0	-3.49	315	329	-14
Nunez, Juan	BAR	1	5	-2	-3.54	325	330	-5
Atamna, Adam	ASV	17	177	8	-3.64	293	331	-38
Mijailovic, Uros	PAR	5	12	-6	-3.97	334	332	+2
Joksimovic, Stefan	BAS	11	65	-5	-5.27	332	333	-1
Cancar, Vlatko	MIL	2	15	-5	-5.40	332	334	-2
Baiocchi, Matteo	VIR	5	50	-6	-7.49	334	335	-1

The player table therefore shows how recorded action credit is redistributed within the same official fields and exposure history. As a separate descriptive check, secondary player associations were small. Across 8,741 positive-minute 2025–26 player-games, raw Spearman correlation with raw plus-minus was 0.3012 for official PIR and 0.3186 for RM-PIR. Partial Spearman correlations controlling possession exposure as a covariate—without dividing either outcome by that exposure—were 0.2795 and 0.2942, a difference of 0.0147 (whole-game bootstrap interval 0.0092–0.0205). Team-margin-residualised on-court plus-minus, defined in Equation 6, is not adjusted plus-minus: it uses the final margin, does not adjust for teammates, opponents, lineups or stints, and remains mechanically dependent on observed plus-minus. Its possession-normalised form shares a denominator with PIR per 100 possessions and is reported only as a sensitivity. Association with the next appearance or the mean of exactly the next three appearances was near zero (official 0.0228/0.0347; RM-PIR 0.0195/0.0300). These are convergent-association diagnostics, not player-level validation; a genuine adjusted-plus-minus fit was infeasible because the analysis-ready data contain no lineup-stint design matrix.

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Table 2. Supplementary Table S2. 2025–26 all-stage team rankings over all 402 completed games (380 regular-season, three play-in, 16 playoff and three Final Four games). GP denotes games played. In contrast to Table 1, which restricts every club to the common 38-game regular season, this table includes all completed competition stages; games played consequently range from 38 to 44 because postseason participation is unequal. Ranks are based on cumulative totals and therefore reflect both per-game performance and postseason exposure. The adjacent per-game columns separate scoring rate from cumulative opportunity. Rank change is official PIR rank minus RM-PIR rank, so positive values indicate upward movement. RM-PIR used coefficients selected and fitted exclusively from 2022–23 through 2024–25.

Team	Code	GP	PIR total	RM total	PIR/game	RM/game	PIR rank	RM rank	Δ rank
Olympiacos	OLY	43	4727	4407.00	109.93	102.49	1	1	0
Real	MAD	44	4560	4305.12	103.64	97.84	2	2	0
Valencia	PAM	44	4550	4264.68	103.41	96.92	3	3	0
Monaco	MCO	43	4424	4126.15	102.88	95.96	4	4	0
Panathinaikos	PAN	44	4294	4045.57	97.59	91.94	5	5	0
Zalgiris	ZAL	42	4114	3859.68	97.95	91.90	6	6	0
Hapoel Tel Aviv	HTA	42	4087	3840.48	97.31	91.44	7	7	0
Dubai	DUB	38	3737	3545.84	98.34	93.31	10	8	+2
Crvena Zvezda	RED	39	3780	3542.76	96.92	90.84	9	9	0
Maccabi	TEL	38	3803	3539.55	100.08	93.15	8	10	-2
Fenerbahce	ULK	43	3720	3512.44	86.51	81.68	12	11	+1
Barcelona	BAR	40	3726	3502.92	93.15	87.57	11	12	-1
Milan	MIL	38	3671	3409.86	96.61	89.73	13	13	0
Baskonia	BAS	38	3626	3362.99	95.42	88.50	14	14	0
Paris	PRS	38	3531	3344.34	92.92	88.01	15	15	0
Virtus	VIR	38	3376	3136.22	88.84	82.53	16	16	0
Partizan	PAR	38	3287	3094.08	86.50	81.42	17	17	0
Bayern	MUN	38	3260	3053.61	85.79	80.36	18	18	0
Anadolu Efes	IST	38	3239	2992.09	85.24	78.74	19	19	0
ASVEL	ASV	38	3052	2825.16	80.32	74.35	20	20	0

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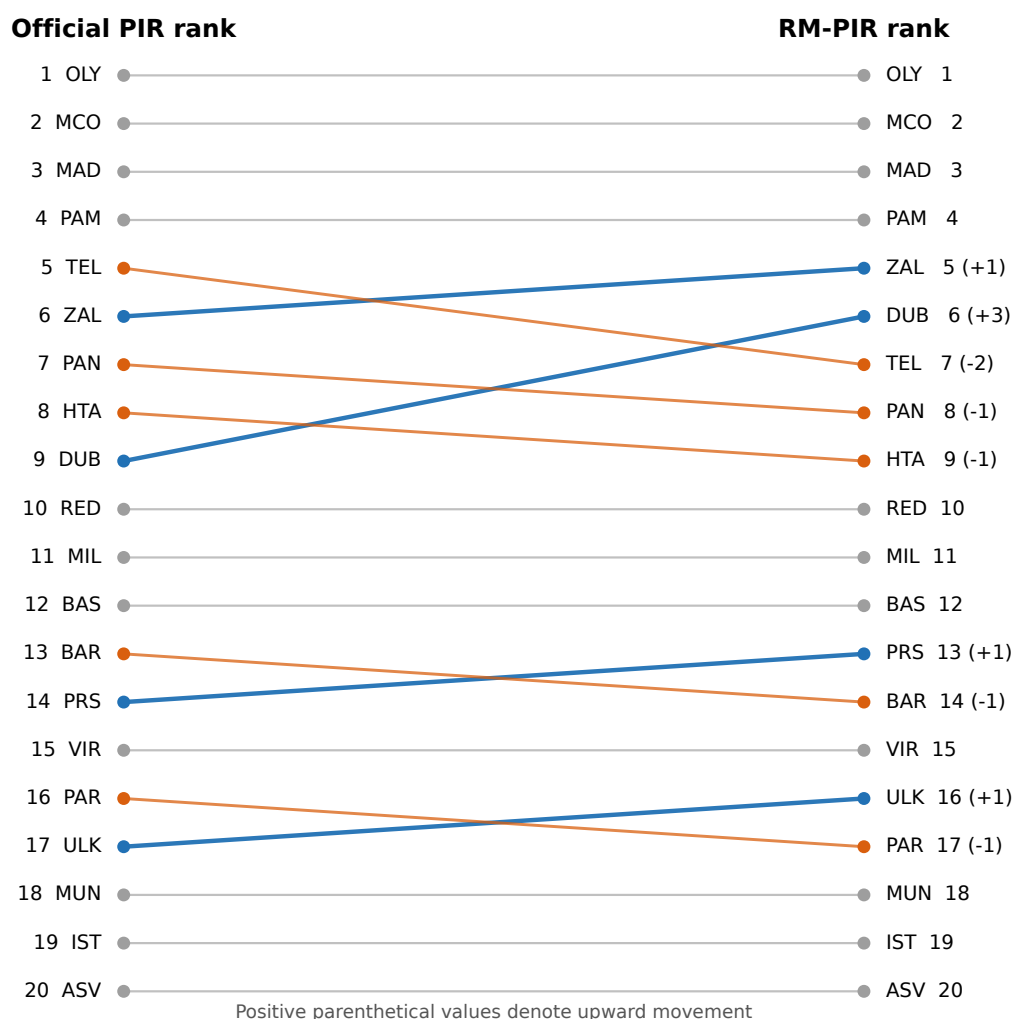


Figure 2. Supplementary Figure S2. 2025–26 regular-season team-rank movement from official PIR to RM-PIR. Rank one is at the top. Blue lines denote upward movement, orange lines downward movement and grey lines unchanged rank. Values reproduce Table 1; positive parenthetical values indicate upward movement.

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